

Assignment - 1

Individual Assignment

Total marks allocated: 10

Task:

Prepare a software architecture document for the following user requirement. You can use the C4 model diagrams to visualize your architecture. Mention the assumptions clearly and justify your architectural decisions (ex: technology selection, containers, how to scale the system etc).

User Requirement:

Colombo International Bookfair, organized by the Sri Lanka Book Publishers' Association, is the largest book fair and exhibition in Sri Lanka. Due to its growing popularity each year, there's an increasing demand among book publishers and vendors to reserve a stall in this exhibition. Therefore, the organizers have decided to build a reservation management system to allow book publishers and vendors to book a suitable stall for their sales in this exhibition.

Book publishers and vendors who wish to reserve a stall at the exhibition should first register themselves via an online portal by providing the necessary information. They should then be navigated to a separate page for stall reservations. Stalls are categorized into three sizes: small, medium, and large based on the dimensions, and are named alphabetically. A map of the exhibition venue should be displayed on this page, with available stalls for reservation on it. Ones that are already reserved should be grayed out. Users are allowed to reserve at most 3 stalls per business.

Once the user selects a suitable stall(s), a pop-up screen should be displayed asking for confirmation for the reservation. Once the reservation is confirmed, an email must be sent to the user confirming the reservation. A unique QR must be generated, which acts as a pass to enter the exhibition premises. This QR should be included in the email to download as well. The user will then be navigated to a home screen. In the home screen, the user is prompted to add the literary genres they will be displaying/selling at the exhibition.

Also, an employee-only portal should be implemented for the organizers to check the availability of stalls and reservations made.

Assume that you are required to build a separate backend Rest API application and separate front-end applications for each portal.